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MOTOR

Abstract

A motor includes: a terminal, including a slit where a winding drawn out from a coil is crimped; and a terminal holder, holding the terminal. The terminal holder includes: an accommodation groove, formed between a first sidewall and a second sidewall arranged at predetermined intervals in a width direction orthogonal to an extension direction of the winding, and accommodating the winding; a rib, provided on a portion of the first sidewall in the extension direction, and reducing a width dimension of the accommodation groove; and a locking protrusion, provided on the second sidewall side at a position offset in the extension direction from the rib, and locking the winding accommodated in the accommodation groove.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Japan application serial no. 2024-030288, filed on Feb. 29, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a motor.

Related Art

[0003] In recent years, efforts have been made to promote the Sustainable Development Goals (hereinafter “SDGs”, the 2030 Agenda for Sustainable Development, adopted at the UN Summit on Sep. 25, 2015). Accordingly, technologies aimed at reducing waste or defective products are known to ensure sustainable production and consumption patterns.

[0004] Conventionally, there has been known a motor which includes: a terminal, in which a lead part of a winding drawn out from a stator coil is crimped, in order to connect the lead part to a substrate; and a terminal holder, holding the terminal. For example, in the motor described in WO 2022/102440, the terminal includes multiple slits in which the lead part is crimped. In the terminal holder, multiple guide grooves (accommodation grooves) are formed to guide the lead part. In a state in which the lead part is temporarily held in each of the multiple guide grooves, the lead part is crimped in the slit of the terminal and the terminal is held by the terminal holder.

[0005] In the guide groove of the terminal holder, a rib is formed on both sides or on one side to temporarily hold the lead part. The lead part is temporarily held in the guide groove by being press-fitted between the ribs, or between a side surface of the guide groove and the rib.

[0006] However, in the motor described above, when the lead part is temporarily held in multiple guide grooves, the lead part must have a press-fit load applied thereto and be inserted into the guide groove. Since the winding is wound on a stator core and formed into a coil while tension is applied thereto by a winding machine, a wire diameter is changed according to the tension applied when the coil is formed. Accordingly, a load required for press-fitting and a push-in amount of the lead part of the winding are changed according to the wire diameter. It is preferable to press-fit the lead part into the guide groove automatically by a machine such as a forming machine. However, in the case where the press-fit load by the machine or the push-in amount is insufficient, manual work by an operator becomes necessary, which is not efficient.

[0007] In the case of attempting to temporarily hold multiple lead parts simultaneously on a single terminal holder, the guide groove may be deformed due to press-fitting of the lead part. In the case where the lead part is press-fitted and a guide groove is increased in width, the guide groove adjacent thereto may be decreased in width. Hence, it may become difficult for the machine to insert the lead part into the guide groove, and it may become necessary to redo the work.

SUMMARY

[0008] A motor of the disclosure includes: a terminal, including a slit where a winding drawn out from a coil is crimped; and a terminal holder, holding the terminal. The terminal holder includes: an accommodation groove, formed between a first sidewall and a second sidewall arranged at predetermined intervals in a width direction orthogonal to an extension direction of the winding, and accommodating the winding; a rib, provided on a portion of the first sidewall in the extension direction, and reducing a width dimension of the accommodation groove; and a locking protrusion, provided on the second sidewall side at a position offset in the extension direction from the rib, and locking the winding accommodated in the accommodation groove.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a longitudinal cross-sectional view of a motor according to an embodiment.

[0010] FIG. 2 is a perspective view of a front surface side showing a configuration of the motor with a rotor removed.

[0011] FIG. 3 is a perspective view of a back surface side showing a configuration of the motor with a driver case and the rotor removed.

[0012] FIG. 4 is an exploded perspective view of the motor.

[0013] FIG. 5 is a perspective view of an area around a terminal and a terminal holder.

[0014] FIG. 6 is a perspective view of the terminal.

[0015] FIG. 7 is a perspective view showing a state before a lead part of a coil is inserted into an accommodation groove of the terminal holder.

[0016] FIG. 8 is an enlarged explanatory diagram of an area around the accommodation groove.

[0017] FIG. 9 is an explanatory diagram showing a dimensional relationship between the accommodation groove, a rib, and a locking protrusion.

[0018] FIG. 10A and FIG. 10B include explanatory diagrams of a state (FIG. 10A) before the lead part of the coil is inserted into the accommodation groove of the terminal holder and a state (FIG. 10B) after insertion, as viewed from a longitudinal direction of the accommodation groove.

[0019] FIG. 11A and FIG. 11B include explanatory diagrams of a state (FIG. 11A) before the lead part of the coil is inserted into the accommodation groove of the terminal holder and a state (FIG. 11B) after insertion, as viewed from a depth direction of the accommodation groove.

[0020] FIG. 12A and FIG. 12B include explanatory diagrams of a state (FIG. 12A) before the terminal is crimped to the lead part and a state (FIG. 12B) after crimping, as viewed from the longitudinal direction of the accommodation groove.

DESCRIPTION OF THE EMBODIMENTS

[0021] The disclosure provides a motor in which temporary holding of a winding can be made easy and work efficiency can be improved.

[0022] According to the disclosure, it is possible to easily temporarily hold the lead part and improve work efficiency and yield rate.

[0023] The following describes, as one aspect of a motor according to an embodiment of the disclosure, a motor which is, for example, mounted in a vehicle such as an automobile and drives a power slide door, a fan device, an electric hydraulic pump, or the like.

[Overall Configuration of Motor 2]

[0024] A configuration of the motor 2 will be described with reference to FIG. 1 and FIG. 2. FIG. 1 is a longitudinal sectional view of the motor 2. FIG. 2 is a perspective view of a front surface side showing the configuration of the motor 2 with the rotor 23 removed.

[0025] As shown in FIG. 1 and FIG. 2, the motor 2 is a so-called “integrated electromechanical type” electric motor that includes a brushless motor 11 of an outer rotor type and a substrate 13 on which a driver circuit 12 is mounted.

[0026] The motor 2 includes, in addition to the brushless motor 11, the driver circuit 12, and the substrate 13, a motor bracket 14, a driver case 15, a terminal holder 16 (see FIG. 4), terminals 17A to 17C (see FIG. 4), and a connector unit 18.

[0027] The brushless motor 11 is supported by the motor bracket 14. The brushless motor 11 is arranged on one side (front surface side) in a thickness direction of the motor bracket 14. The driver case 15 is fastened to the other side (back surface side) in the thickness direction of the motor bracket 14 by multiple screws.

[0028] The connector unit 18 formed by integrating two connectors to which an external harness is connected is attached to an end of the motor bracket 14. The brushless motor 11, the driver circuit

12, and the connector unit **18** are electrically connected.

[0029] As shown in FIG. **1**, the brushless motor **11** includes a shaft **21**, multiple bearings **22** provided on an outer periphery of the shaft **21**, a rotor **23** rotatably supported around an axis of the shaft **21** via the bearing **22**, and a stator **24** of an annular shape fixed with a predetermined interval in a radial direction from the rotor **23**.

[0030] The shaft **21** is a fixing shaft fixed to the front surface side of the motor bracket **14**. In the following description related to the components of the motor **2**, an axial direction of the shaft **21** is simply referred to as the “axial direction”, a radial direction centered on the axis of the shaft **21** is simply referred to as the “radial direction”, and a circumferential direction centered on the axis of the shaft **21** is simply referred to as the “circumferential direction”.

[0031] The rotor **23** includes multiple permanent magnets **231** arranged at equal intervals in the circumferential direction to surround an outer periphery of the stator **24**, and a rotor yoke **232** that supports the multiple permanent magnets **231** and is rotatably supported on the shaft **21**.

[0032] The rotor yoke **232** is arranged on the front surface side of the motor bracket **14** to be concentric with the axis of the shaft **21**. The rotor yoke **232** is rotatably supported on the shaft **21** via multiple bearings **22**. Furthermore, the rotor yoke **232** includes an outer peripheral wall **232A**, an inner peripheral wall **232B**, and a connecting wall **232C**.

[0033] The outer peripheral wall **232A** presents a cylindrical outer shape. The outer peripheral wall **232A** is arranged radially outside of the stator **24**. Furthermore, the outer peripheral wall **232A** supports the multiple permanent magnets **231** on its inner peripheral surface. In other words, the multiple permanent magnets **231** are fixed to the inner peripheral surface of the outer peripheral wall **232A** at predetermined intervals in the circumferential direction.

[0034] The inner peripheral wall **232B** presents a cylindrical outer shape. The inner peripheral wall **232B** is arranged radially inside of the stator **24**. Furthermore, the inner peripheral wall **232B** is rotatably supported on the shaft **21** via multiple bearings **22**.

[0035] The connecting wall **232C** presents a disk-like outer shape. The connecting wall **232C** connects one axial end of the outer peripheral wall **232A** and one axial end of the inner peripheral wall **232B**. Furthermore, the connecting wall **232C** is arranged on a side opposite to the motor bracket **14** with the stator **24** therebetween. The connecting wall **232C** is arranged to face and be spaced by a predetermined interval in the axial direction from the stator **24**.

[Configuration of Stator **24**]

[0036] The stator **24** is accommodated in a space surrounded by the outer peripheral wall **232A**, the inner peripheral wall **232B**, the connecting wall **232C**, and a front surface of the motor bracket **14**. The stator **24** is fixed on the front surface side of the motor bracket **14**, radially inside of the multiple permanent magnets **231**. Furthermore, the stator **24** faces and is spaced by a predetermined gap in the radial direction from the multiple permanent magnets **231**.

[0037] As shown in FIG. **2**, the stator **24** includes a stator core **241** of a cylindrical shape, an insulating stator insulator **242** mounted on both axial sides of multiple teeth protruding radially outward from the stator core **241**, and a coil **243** in which a winding is wound on the stator insulator **242**. The coil **243** is a conductive winding, and is, for example, formed from a copper wire.

[0038] The stator **24** generates a magnetic field by a current flowing through the coil **243**. By attraction force and repulsion force generated between the magnetic field generated by the coil **243** and the multiple permanent magnets **231**, the rotor yoke **232** is rotated about the axis of the shaft **21**.

[0039] The driver circuit **12** controls the generation of magnetic fields by multiple coils **243**. The driver circuit **12** is composed of multiple electronic parts (for example, transistors, diodes, and resistors) surface-mounted on a front surface of the substrate **13** facing the motor bracket **14**. The electronic parts constituting the driver circuit **12** are not arranged on a back surface side of the substrate **13** facing the driver case **15**.

[Configuration of Motor Bracket **14**]

[0040] FIG. **3** is a perspective view of a back surface side showing a configuration of the motor **2** with the driver case **15** and the rotor **23** removed. FIG. **4** is an exploded perspective view of component parts arranged on the back surface side of the motor bracket **14**.

[0041] As shown in FIG. **3** and FIG. **4**, the substrate **13** is fastened to the motor bracket **14** by multiple screws **25**. On the back surface side (side facing the substrate **13**) of the motor bracket **14**, the terminal holder **16** is provided. The motor bracket **14** includes an opening **14A**. The opening **14A** penetrates from the front surface side to the back surface side of the motor bracket **14**, allowing a lead part **244** of the winding forming the coil **243** to be drawn out to the back surface side of the motor bracket **14**. The lead part **244** has elastic force and is elastically deformed when inserted into an accommodation groove **36** to be described later.

[Configuration of Terminals **17A** to **17C**]

[0042] The terminals **17A** to **17C** electrically connect the driver circuit **12** and the coil **243**. The brushless motor **11** includes the three terminals **17A** to **17C** in order to supply three-phase (U-phase, V-phase, and W-phase) power to multiple coils **243**. That is, the brushless motor **11** is a three-phase AC motor. In the present embodiment, the brushless motor **11** includes a total of 12 coils **243**, with 4 coils for each phase.

[0043] As shown in FIG. **5**, the terminals **17A** to **17C** are made of conductive metal and are integrally formed with a main body **26** and a terminal part **27**. The terminals **17A** to **17C** are held by holding parts **16A** to **16C** of the terminal holder **16**.

[0044] As shown in FIG. **6**, the main body **26** is formed to have a substantially horizontal U-shape in cross-section and includes two fitting pieces **26A** and **26B**. The fitting pieces **26A** and **26B** are fitted with fitting grooves **37A** and **37B** (see FIG. **7**) of the terminal holder **16** to be described later. The fitting piece **26A** and the fitting piece **26B** are arranged parallel to each other. FIG. **6** illustrates a configuration of the terminal **17A**, and the terminals **17B** and **17C** have the same configuration as the terminal **17A**.

[0045] The fitting piece **26A** and the fitting piece **26B** respectively include four slits **28A** and four slits **28B**. In the slits **28A** and **28B**, the lead part **244** of the winding drawn out from the coil **243** is crimped. The slits **28A** and **28B** each include a crimping part **28C** where the lead part **244** is crimped, and a tapered part **28D** that widens in width from the crimping part **28C** toward a tip in a crimping direction. The crimping part **28C** has a width smaller than a wire diameter of the lead part **244**. The tapered part **28D** has a width larger than the wire diameter of the lead part **244** and is formed in a shape to guide the lead part **244** into the crimping part **28C**.

[0046] The terminal part **27** includes multiple connection pieces **29** and is electrically connected to the driver circuit **12**. The substrate **13** includes a through hole **13A** (see FIG. **4**) and a land **13B** (see FIG. **4**). The through hole **13A** penetrates from the front surface (mounting surface) to the back surface of the substrate **13**. The land **13B** is a conductive film continuously formed from an inner wall surface of the through hole **13A** to around the through hole **13A**, and is connected to a pattern of an output terminal for each phase in the driver circuit **12**. The land **13B** is soldered by the connection piece **29** inserted into the through hole **13A**. Accordingly, the terminals **17A** to **17C** are connected to the output terminals for each phase in the driver circuit **12**.

[Configuration of Terminal Holder **16**]

[0047] As shown in FIG. **5**, the terminal holder **16** includes the holding parts **16A** to **16C** and an opening **16D**. The terminal holder **16** is made of an insulating material (for example, resin). The holding parts **16A** to **16C** hold the terminals **17A** to **17C** respectively, and also hold the lead part **244** of the winding crimped in the slits **28A** and **28B** of the terminals **17A** to **17C**. The opening **16D** penetrates from a front surface side to a back surface side of the terminal holder **16**, allowing the lead part **244** of the winding forming the coil **243** to be drawn out to the back surface side of the terminal holder **16**.

[0048] As shown in FIG. **7**, in the holding part **16A**, multiple accommodation grooves **36**, the

fitting grooves **37A** and **37B**, multiple ribs **38**, and multiple locking protrusions **39** are formed. The accommodation groove **36** accommodates the lead part **244** of the winding drawn out from the coil **243**. In the present embodiment, the holding part **16A** includes four each of the accommodation grooves **36**, multiple ribs **38**, and locking protrusions **39**, and holds four lead parts **244**. The disclosure is not limited thereto, and the number of lead parts **244** held by one holding part can be appropriately changed according to the number of coils **243** and the number of phases of power supplied to the coil **243**. Accordingly, the number of accommodation grooves **36**, multiple ribs **38**, locking protrusions **39**, and slits in the terminals **17A** to **17C** can also be changed.

[0049] FIG. 7 illustrates only the configuration of the holding part **16A**, and the holding parts **16B** and **16C** have similar configurations. In FIG. 7 and FIG. 9, only one lead part **244** is illustrated. However, in an actual process, it is preferable to simultaneously insert multiple lead parts **244** into multiple accommodation grooves **36**. The accommodation groove **36** is an elongated groove whose longitudinal direction is an extension direction Z of the lead part **244** accommodated therein. Hereinafter, the extension direction of the lead part **244** accommodated in the accommodation groove **36** is simply referred to as the “extension direction”, a width direction of the accommodation groove **36** orthogonal to the extension direction is simply referred to as the “width direction”, and a depth direction of the accommodation groove **36** orthogonal to both the extension direction and the width direction is simply referred to as the “depth direction”. The four accommodation grooves **36** are arranged in parallel with respect to the width direction X.

[0050] The fitting grooves **37A** and **37B** are arranged in positions intersecting the extension direction Z of the accommodation groove **36** and are formed along the width direction X. In the present embodiment, the fitting grooves **37A** and **37B** are arranged near both ends of the accommodation groove **36**.

[0051] The fitting grooves **37A** and **37B** are formed with dimensions in the width direction X that match the dimensions of the fitting pieces **26A** and **26B** of the terminal **17A**. Accordingly, when the lead part **244** is crimped into the slits **28A** and **28B**, since the fitting pieces **26A** and **26B** are simultaneously fitted with the fitting grooves **37A** and **37B**, the terminal **17A** can be held by the holding part **16A**.

[0052] As shown in FIG. 8, the accommodation groove **36** is formed between a first sidewall **36A** and a second sidewall **36B** that are arranged at predetermined intervals in the width direction X. The accommodation groove **36** is a groove having a substantially V-shape in cross-section, having a large width on an upper surface (open end) side and a small width on a bottom surface side.

[0053] Each rib **38** is arranged inside the accommodation groove **36**. The rib **38** is provided on a portion of the first sidewall **36A** in the extension direction Z. Furthermore, the rib **38** protrudes in the width direction X from the first sidewall **36A** toward the second sidewall **36B**. That is, the rib **38** reduces the dimension of the accommodation groove **36** in the width direction X at a portion in the extension direction Z.

[0054] The locking protrusion **39** is provided on the second sidewall **36B** side at a position offset in the extension direction Z from the rib **38**. The locking protrusion **39** protrudes in a direction (namely, from the second sidewall **36B** side toward the first sidewall **36A** side; width direction X) opposite to a direction in which the rib **38** protrudes. Furthermore, in the depth direction Y, the locking protrusion **39** is separated from the bottom surface of the accommodation groove **36**. In other words, a space accommodating the lead part **244** of the winding is formed between the bottom surface of the accommodation groove **36** and the locking protrusion **39**. Accordingly, the locking protrusion **39** locks the lead part **244** of the winding accommodated in the accommodation groove **36**.

[0055] The lead part **244** accommodated inside the accommodation groove **36** is guided by the rib **38** to the side where the locking protrusion **39** is located. Furthermore, the locking protrusion **39** locks and restricts the lead part **244** from detaching from the accommodation groove **36**. Hence, the lead part **244** is temporarily held inside the accommodation groove **36**. Temporary holding refers to

a state in which, while the lead part **244** is not fixed to the accommodation groove **36**, the lead part **244** does not detach from the accommodation groove **36**.

[0056] As described later, when the lead part **244** is inserted into the accommodation groove **36**, the locking protrusion **39** always comes into contact with the lead part **244**. However, since the locking protrusion **39** has a tip formed in a curved surface shape, damage can be prevented from being caused to the lead part **244**. The locking protrusion **39** has rigidity and is not elastically deformed when locking the lead part **244**.

[0057] As shown in FIG. **9**, in the width direction X, an interval D1 between the rib **38** and the second sidewall **36B** facing the rib **38** is larger than a wire diameter R1 of the lead part **244**. In the width direction X, an interval D2 between the locking protrusion **39** and the first sidewall **36A** facing the locking protrusion **39** is larger than the wire diameter R1 of the lead part **244**. In the width direction X, an interval D3 between a tip of the rib **38** and the tip of the locking protrusion **39** is smaller than the wire diameter R1 of the lead part **244**.

[0058] In the depth direction Y, the rib **38** is located closer to the bottom surface side of the accommodation groove **36** than the locking protrusion **39**. Being located on the bottom surface side here means that at least a portion of the rib **38** is located closer to the bottom surface side of the accommodation groove **36** than the locking protrusion **39** in the depth direction Y. Accordingly, in the case where the lead part **244** is inserted into the accommodation groove **36**, since the lead part **244** is guided by the rib **38** to the side where the locking protrusion **39** is located and to the lower side (bottom surface side) of the locking protrusion **39**, the lead part **244** is reliably temporarily held inside the accommodation groove **36**.

[0059] The rib **38** includes a flat surface **38A** and an inclined surface **38B**. The flat surface **38A** is located on the bottom surface side of the accommodation groove **36** and is the most protruding surface of the rib **38** in the width direction X. The inclined surface **38B** is inclined from the first sidewall **36A** toward the direction in which the rib **38** protrudes. Accordingly, in the case where the lead part **244** is accommodated inside the accommodation groove **36**, the lead part **244** is guided to the locking protrusion **39** side by the inclined surface **38B**. Hence, the lead part **244** is further reliably temporarily held inside the accommodation groove **36**.

[Assembly Process of Motor 2]

[0060] Next, a process for assembling the substrate **13**, motor bracket **14**, terminal holder **16**, and terminals **17A** to **17C** will be described. In the following description, a process in which the terminal **17A** and the lead part **244** are held by the holding part **16A** of the terminal holder **16** will be mainly described. However, a process for holding the terminals **17B**, **17C** and the lead part **244** by the holding parts **16B** and **16C** is performed in the same manner, and a description thereof will thus be omitted.

[0061] First, the terminal holder **16** is attached to the back surface of the motor bracket **14**. The terminal holder **16** is fastened to the motor bracket **14** by multiple screws. During the process for attaching the terminal holder **16** to the motor bracket **14**, the lead part **244** of the winding forming the coil **243** is drawn out to the back surface side of the terminal holder **16** through the openings **14A** and **16D** (see FIG. **4** and FIG. **5**).

[0062] Multiple (in the present embodiment, four) lead parts **244** of the windings drawn out from the coil **243** are inserted into the accommodation groove **36** of the holding part **16A**. The insertion of the lead parts **244** into the accommodation groove **36** is performed automatically by a machine such as a forming machine. FIG. **10A** and FIG. **12A** show a state during the insertion of the lead parts **244** into the accommodation groove **36**.

[0063] Inside the accommodation groove **36**, the rib **38** that reduces the dimension in the width direction X and the locking protrusion **39** protruding on the side opposite to the rib **38** are arranged. Thus, the lead part **244** comes into contact with the rib **38** and the locking protrusion **39**.

Accordingly, the lead part **244** is inserted into the accommodation groove **36** while bending into an S-shape due to elastic force.

[0064] As shown in FIG. 10B and FIG. 11B, after the lead part **244** has climbed over the locking protrusion **39**, the lead part **244** is restored in shape due to its elastic force. That is, the lead part **244** is in a linear shape and is accommodated inside the accommodation groove **36**.

[0065] It is preferable to change a positional relationship between the rib **38** and the locking protrusion **39** according to rigidity of the lead part **244**. In the case where the lead part **244** has high rigidity (low elastic force), the lead part **244** is less likely to bend into an S-shape. Accordingly, it is preferable that a distance L1 (see FIG. 11A) between the rib **38** and the locking protrusion **39** in the extension direction Z be appropriately changed according to the rigidity of the lead part **244**.

Accordingly, the lead part **244** becomes likely to bend into an S-shape.

[0066] After the lead part **244** has climbed over the locking protrusion **39**, the lead part **244** is in a linear shape and is accommodated inside the accommodation groove **36**. In the case where the lead part **244** is accommodated inside the accommodation groove **36**, the locking protrusion **39** locks and restricts the lead part **244** from detaching from the accommodation groove **36**.

[0067] As described above, by inserting the lead part **244** into the accommodation groove **36** with a small load, since the lead part **244** is temporarily held by the rib **38** and the locking protrusion **39**, the lead part **244** can be temporarily held by simple work. In particular, when the lead part **244** climbs over the locking protrusion **39**, the lead part **244** is deformed into an S-shape. This deformation is due to the elastic force of the lead part **244**, and the lead part **244** can be inserted into the accommodation groove **36** with a small load.

[0068] As described above, in the width direction X, the interval D1 between the rib **38** and a side surface of the accommodation groove **36** facing the rib **38** as well as the interval D2 between the locking protrusion **39** and a side surface of the accommodation groove **36** facing the locking protrusion **39** are larger than the wire diameter R1 of the lead part **244**. Hence, in the process for inserting the lead part **244** into the accommodation groove **36**, less load is required than in the case of press-fitting the lead part, and the lead part **244** can be easily temporarily held by simply inserting the lead part **244** into the accommodation groove **36** with a small load and the same push-in amount.

[0069] As shown in FIG. 12A, after the lead part **244** is temporarily held in the accommodation groove **36**, the lead part **244** is crimped into the slits **28A** and **28B** of the terminal **17A**. As shown in FIG. 12B, while the lead part **244** is crimped into the slits **28A** and **28B**, the fitting pieces **26A** and **26B** are simultaneously fitted into the fitting grooves **37A** and **37B**. Accordingly, the terminal **17A** and the lead part **244** can be held by the holding part **16A**.

[0070] In the disclosure, unlike a conventional motor, the temporary holding of the lead part in the accommodation groove is not achieved by press-fitting. Furthermore, based on the above-described dimensional relationship between the wire diameter of the lead part **244** and the accommodation groove **36**, rib **38**, and locking protrusion **39**, in the temporarily held state, the lead part **244** has a gap with respect to the accommodation groove **36**. That is, the lead part **244** is movable in the width direction X within the accommodation groove **36**. Hence, when crimping is performed in the slits **28A** and **28B**, the lead part **244** is able to follow the tapered part **28D** of the slits **28A** and **28B** and smoothly enter the crimping part **28C**.

[0071] When the lead part **244** is crimped into the slits **28A** and **28B**, coating on its surface is peeled off. Accordingly, the terminals **17A** to **17C** are electrically connected with the lead part **244**. After the terminals **17A** to **17C** are connected with the lead part **244**, the substrate **13** is fastened to the motor bracket **14** by the screw **25**. Furthermore, the land **13B** of the substrate **13** is soldered to the connecting piece **29** of the terminals **17A** to **17C**. Accordingly, the driver circuit **12** and the coil **243** are electrically connected via the terminals **17A** to **17C**.

[0072] As described above, in the process for inserting the lead part **244** into the accommodation groove **36**, the lead part **244** can be easily temporarily held by simple insertion with a constantly small load and the same push-in amount. Thus, these processes can be automatically performed by a machine. That is, manual work by an operator becomes unnecessary, and work efficiency is

improved.

[0073] In the disclosure, the temporary holding of the lead part **244** in the accommodation groove **36** is not achieved by press-fitting. That is, even if the process for inserting the lead part **244** into the accommodation groove **36** is performed, since the accommodation groove **36**, rib **38**, and locking protrusion **39** are not deformed, it will not be difficult to insert the lead part **244** into the accommodation groove **36** as in a conventional motor. Thus, in the motor **2** of the disclosure, multiple lead parts **244** can be simultaneously inserted into multiple accommodation grooves **36**. Since the lead part **244** can be inserted into the accommodation groove **36** with a small load and the lead part **244** can be temporarily held by the rib **38** and the locking protrusion **39**, the lead part **244** will not detach from the accommodation groove **36**. That is, it is possible to stably maintain the lead part **244** in a temporarily held state, and the yield rate is improved.

[0074] The above-described embodiments are examples for describing the disclosure and are not intended to limit the scope of the disclosure to only those embodiments. Those skilled in the art can make appropriate modifications within the scope of the disclosure without departing from the spirit of the disclosure.

Claims

1. A motor, comprising: a terminal, comprising a slit where a winding drawn out from a coil is crimped; and a terminal holder, holding the terminal, wherein the terminal holder comprises: an accommodation groove, formed between a first sidewall and a second sidewall arranged at predetermined intervals in a width direction orthogonal to an extension direction of the winding, and accommodating the winding; a rib, provided on a portion of the first sidewall in the extension direction, and reducing a width dimension of the accommodation groove; and a locking protrusion, provided on the second sidewall side at a position offset in the extension direction from the rib, and locking the winding accommodated in the accommodation groove.
 2. The motor as claimed in claim 1, wherein, in the width direction, an interval between the rib and the second sidewall is larger than a wire diameter of the winding.
 3. The motor as claimed in claim 2, wherein, in the width direction, an interval between the locking protrusion and the first sidewall is larger than the wire diameter of the winding.
 4. The motor as claimed in claim 3, wherein, in the width direction, an interval between a tip of the rib and a tip of the locking protrusion is smaller than the wire diameter of the winding.
 5. The motor as claimed in claim 1, wherein, in a depth direction of the accommodation groove that is orthogonal to the extension direction and the width direction, the rib is located closer to a bottom surface of the accommodation groove than the locking protrusion.
 6. The motor as claimed in claim 1, wherein the rib comprises an inclined surface that guides the winding accommodated in the accommodation groove toward the locking protrusion side.
 7. The motor as claimed in claim 1, wherein the terminal holder comprises a plurality of the accommodation grooves, and the plurality of the accommodation grooves are arranged in parallel with respect to the width direction.
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